Thermal effects and energy budgets

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a temperature change only after stating how much energy enters the material.

SUCCESS CRITERIA - I can...

1. Apply an explicit absorbed-energy fraction.
2. Use mass and specific heat in a temperature estimate.
3. State the no-phase-change and no-loss assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “absorbed” and “specific heat” to explain your model or result. Sentence starters are optional.

absorbed: taken into the specified object or system

specific heat: energy per mass needed for a one-degree temperature change

energy budget: an account of all energy stored and transferred

01 Bring back an earlier idea

A refrigerator uses 15 J electrical work while moving heat from its colder interior to a warmer room.
Explain why this is not spontaneous cold-to-hot heat flow.

Word help: spontaneous = able to occur without the specified external work input

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $Q_{absorbed} = mc\Delta T$; $\Delta T = Q_{absorbed}/(mc)$.
First identify the absorbed energy. Use only when there is no phase change and c is approximately constant.

Worked reasoning: A surface receives 500 J. A model accounts for 200 J reflected, 250 J absorbed as heating and 50 J transmitted. The full account sums to 500 J.

Your turn: Draw proportional energy arrows and identify which part can directly raise this surface's temperature.

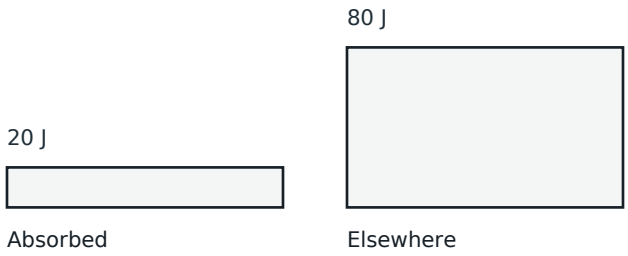
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

A model releases 100 J. Only 20% enters a sample.
Draw the energy partition and calculate energy absorbed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Build the solution

The sample has m=1 kg and c=10 J/(kg·°C). With 20 J absorbed, find$\Delta T = 20 \div (1 \times 10)$.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A temperature prediction exceeds a material’s melting point, but the calculation used only mcΔT. What must be added or reconsidered?

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Choose a method

A device receives 60 J and delivers 42 J useful mechanical energy.
Find other energy and the useful fraction.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

If absorbed energy doubles for the same mass and material with no phase change, compare temperature rises.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two surfaces absorb 250 J but have different masses of the same material.
Compare their temperature rises if one mass is twice the other.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3